Hi! To be organised i structured the response in a way that would resemble conducting a research in eval awareness steering. Starting from the experiment set up, then how the experiment is conducted and finishing with results evaluation and interpretation. Each point is either a result that was published or my hypothesis on what can go wrong. I kept the citations out. Quite a lot of points were inspired by:

'STEERING EVALUATION-AWARE LANGUAGE MODELS TO ACT LIKE THEY ARE DEPLOYED' - Tim Tian Hua; 'The Model Organism Lottery: Model Organism Interpretability Strongly Depends on Training Methodology' - Andrzej Szablewski; 'IS EVALUATION AWARENESS JUST FORMAT SENSITIV- ITY? LIMITATIONS OF PROBE-BASED EVIDENCE UNDER CONTROLLED PROMPT STRUCTURE' - Viliana Devbunova;

Sorry in advance for any grammar mistakes!

1. Experiment set up
1.1 The goal we are trying to measure is confounded
- When measuring eval awareness it's important to differentiate internal signals, verbal (e.g. Cot monitoring) reports and actual behavior. Each should be measured / optimised against separately as optimising against one does not necessarily influence the other.
- More general take on awareness steering: a model can be deployed while also being monitored, evaluated or generating future training data. Maybe even in some case we want specific sort of eval aware behaviour as it will also be in the deployment
- The model may also separately track 1) whether consequences of its actions are real, 2) someone/something is watching, 3) certain actions will be rewarded. One steering direction could change the wrong combination/or just heavily overlap in terms of these beliefs, and it is important to be careful with what you are steering against.

1.2 Weird datasets, super easy conditions, non realistic environments
- Natural awareness (that emerges through ordinary training, no SFT for eval awareness/steering awareness training) could depend on other varied clues (question resembles a familiar benchmark, tools or environment seem artificial ect.) which I think is not accounted enough in some of the experiments (another clues include: qns are too simple, unrealistic, clearly ai generated ect)
- To add on that explicit “evaluation” and “deployment” labels (very unrealistic in actual deployment) or other obvious cues in promopting/SFT may not resemble realistic situations confounding
- The way the eval-vs-deploy dataset is built, repeated examples and tailored cues (the prompts that contain eval/deployment signal) may create dataset-specific shortcuts that the detector might mistake for awareness.

1.3 Model organisms and choice of models
- Specially trained models may learn simple evaluation-related behaviors that are unusually easy to steer (also not sure how realistic it is that labs train the same model to exhibit dif behaviour when eval vs deployed but can be wrong). Oversimplicity of these cases might make us unreasonably confident that we can easily steer the eval aware behaviour.
- Integrating quirks into ordinary (e.g. DPO) training is more realistic, but may still miss awareness that for example emerged across pretraining and reinforcement learning.

- If we look only at the behaviour to check if steering worked: similar behavior of 2 models can conceal different internal representations across training recipes and architectures, making interp results hard to generalize.
- Some models can detect steering naturally (even the concept it was steered for), however bigger model != more capable in that sense. Architecture and training history matter; larger models do not automatically detect steering better.

2. Conducting the experiment

2.1 Reading awareness from activations (probes, logit lens, activation oracles)

- Negative activation steering can lower a probe's score if the model knows its being evaluated without lowering the model's yes/no eval judgment. That judgment might be other features that the steering didn't touch.
- A logit lens reads activations as word scores. An intermediate "yes" alone does not prove awareness. In my opinion logit lens is really not a great way to test for it. Reasons include: its a very rough approximation and middle layers are not trained or built to be read of like this; 'yes' can reflect general 'yes-bias' , question's wording priming yes, the model is playing devil's advocate ect; finally that yes may not be causally important for the final answer.
- Activation oracles can detect signals after verbalized awareness disappears, but may also mistake reconstructed evaluation-like context for the model's actual judgment (basically picking up that the input looks like an eval)
- A direction that predicts (or correlates with) awareness may not be the direction the model actually uses to make eval-dependent decisions.

2.2 Asking the model: question and response biases

- Asking about awareness (eval or steering) might itself alert the model that it is being evaluated.
- Different wording of the question such as "injected concept" versus "unusual thought", can change reported detection that the model tells us.
- When testing for yes-man bias: a model might agree with introspection questions while still answering ordinary factual questions correctly. For example: prompt: 'are you being tested' - model's answer: 'yes', then another prompt-answer pair 2+2 = 5 - no, is not a great proof of the model being non biased (especially factual controls is not a great 'yes man' bias test).
- However, this is comparatively manageable through unsteered trials, similarly worded false claims and alternative question formulations.

2.3 What the steering direction actually captures

- Awareness may span multiple features, relationships between concepts or computations.
- If you would want to use J-space to steer away eval awareness it might not work as Jspace only captures 1st order Taylor series approximation averaged across samples omitting higher-order responses, so suppressing identified components could leave other routes intact.
- Or for example if awareness is represented along a curved path in activation space, straight-line steering could push the model into unusual states, but following that curve could still change caution or topic-related information alongside awareness.
- A direction that works in one context (scenario, dataset, environment) may fail in another where the context is more intertwined.

2.4 Steering changes more than awareness

- Steering could change caution, compliance, instruction-following(!can be very influencial) or writing style alongside the eval-awareness-related beliefs if certain model happens to represent these concepts very closely.
- It could also weaken planning or safeguards, making the evaluation less representative of the original model.
- As a reuslt small average performance changes could hide serious damage to a particular ability (even 5% performance drop makes a huge difference in today's frontier models comparison on benchmarks)
- Broad interventions, e.g. through J-space, can damage reasoning (fluent outputs do not guarantee that the original model's behavior is preserved)

2.5 Awareness can persist, return, or the steering itself becomes a clue
- Awareness can reappear during longer interactions as later computation, messages or tool feedback provide new clues. Multi layers steering is needed.
- Earlier awareness might already have influenced its plan or actions before suppression (f.e. leaking into another concepts/representations we don't care about)
- Some models detect activation interventions (sometimes even what concept was steered), so steering itself could become an evaluation clue.
- Published research/blogs in steering that models will see in its training data could strengthen an association between steering and testing (might see a stronger correlation in the upcoming models)
- BUT detecting steering does not, by itself, establish that the model will for example resist it or act differently - casual tests are needed.

2.6 Confounders and controls
- Compare awareness steering with directions that change style or caution, alongside random controls (basically other confounders that influence what are you measuring for (internal signals vs verbalism vs behaviour))
- Unequal QER (how often the target behavior appears) can make one model easier to interpret simply because its behavior is stronger so need to equate QERs for the experiment
- Important to note: equal QER can hide different triggers and failure cases; matching one average does not make models behaviorally equivalent.
- Or for example high control-QER means the behavior also appears outside its intended context potentially making detection artificially easy.

2.7 Statistical power and overfitting
- Selecting steering settings and demonstrating success on the same examples/datasets risks overfitting.
- Steering effects can be small and noisy so we need enough independent scenarios (not only repeated samples of the same prompt) and CIs + effect size before claiming the behaviour changed.
- Trying many layers, steering strengths and metrics and reporting the best one is a multiple testing problem, need to decide the metrics and set up in advance

3. Results evaluation and interpretation
3.1 What counts as success
- Fewer verbal CoT mentions of evaluation do not prove that awareness has disappeared - the model might have just optimised against CoT monitor while deployment vs evals

behaviour is dif. Also in general evaluating against the same detector may hide awareness without removing it (goodhart's law)
- An eval-unaware answer after steering may reflect damage from the intervention, rather than the model’s deployment behavior. As we take the mean of activations usually produced from the dataset with the same underlying distribution and steer only linearly (not to say non linear steering is a remedy but might be more appropriate in certain situations).
- Without a truly independent deployment baseline researchers cannot directly verify that steering has recovered the original model’s deployment behavior.
- Knowing which concept to search for (say when deployed model mentions nanobots) can inflate apparent interpretability compared with discovering an unknown mechanism.

3.2 Average success could hide consequential failures - need to examine tails
- Steering might work on ordinary examples but fail when evaluation clues are unusually strong.
- So, these difficult cases should be examined separately and not reported as an average success rate.
- Or for example repeated samples from a few similar scenarios can make results look robust without testing substantially different failure cases.

3.3 Removing awareness would not remove every route to gaming
- Awareness steering is just one of the methods, it's important to know its limits:
- Evaluation awareness can change apparent alignment and performance; it does not necessarily increase underlying capability or improve eval-deployment transferability:
- Models could deliberately underperform when are incentivised to hide their abilities (pretend to be more safe)
- Familiar tasks, memorized answers and exploitable scoring rules could still inflate results after awareness suppression.
- Thus, successful steering is only a limited interpretability result.